DATA 305: Designing AI Agents
Fall 2026 Syllabus

About this course

What is an AI Agent? How is it different from a Chatbot? Are Agents everywhere? Are they safe to use? In this class, we will address these questions, working up to a fundamental understanding of what AI Agents are, how they operate, and how to use them responsibly including in secure environments. We will start with the brain of Agents, Large Language Models (LLMs), demystifying how LLMs work and exploring how their probabilistic nature influences the outputs they produce and ultimately Agent behavior. Tool calling is fundamental to both modern LLMs and Agents, allowing them to move past their training and interact with external resources. We will explore different types of tools that can be used, how to create tool calls using code, and the inherent risks and responsibilities. AI Agents can make decisions autonomously, and may need memory for optimal decision-making. We will discuss the use of Retrieval-Augmented Generation (RAG) systems, both for LLMs and Agentic RAG and how they differ. Finally, we will explore how Agents can work together, both synergistically and adversarially in the context of multi-Agent workflows. Throughout the semester, we will focus on problem-solving skills and process, brainstorming novel solutions together.

Prerequisite: (DATA 101 or CSCI/DATA 140 or 141) and DATA 146/201
Credit Hours: 3
Course Meeting Information: S02 – MWF 1200 to 1250 ISC 3349; S03 – MWF 1400 to 1450 ISC 2363

Course Piazza Page: https://piazza.com/wm/fall2026/data305/home

Instructional Staff Information

Dana Willner, dlwillner@wm.edu – preferred method of contact is private post on Piazza
Office hours:

- In-person: W 1630-1730 (please give me the ten minutes after my last class ☺) ISC 3319, F 1000-1100 (hard stop at 11 AM) ISC 3319
- Zoom: T 1100-1200, R 1600-1700 and by appointment: https://cwm.zoom.us/j/85357678971
- Likely some additional hours added ad hoc

Learning objectives

- Differentiate between Chat Bots, AI Assistants, and AI Agents in order to identify which should be used to accomplish a particular system design or goal
- Understand the probabilistic nature of LLMs and their relationship to model hyperparameters in order to tune models for optimal performance and understand varying outputs
- Build asynchronous workflows with validated outputs to create Agentic Applications and Agents
- Compare and contrast Agentic Design Patterns
- Deploy built-in tools and custom functions in order to enhance the capabilities of LLMs and Agents
- Create Agents using industry standard Agentic Frameworks
- Explain the risks inherent in the use of LLMs and Agents in order to choose systems that maintain accountability and security in a business setting
- Understand how RAG systems can establish long-term memory
- Develop processes and solutions to real-world problems in order to develop critical thinking skills

Course Modules

In order to achieve our learning objectives, course materials are organized into module of related content.
Module 0: Introduction
Module 1: LLM Fundamentals
Module 2: Agentic Design
Module 3: Tool Calling
Module 4: Memory, RAG, and Intro to Multi-Agent Workflows

Important Dates

Add/Drop Deadline: September 4, 2026
Withdrawal Deadline: October 26, 2026
Final Exam (Project) Submission Deadline: Tuesday December 8, 1700

Method of Evaluation

For this course evaluation will involve both in-class (engagement and assessments) and outside of class (assignments) activities. The purpose of on-paper in-class assessments is to prepare you for job interviews where you may need to recall information quickly with limited resources…they aren't a trick, and you will see below that you will be able to bring notes for all of them.

Engagement and In-Class Activities (120 points total/12%)

You are an important and valued member of our class, and your engagement and insights are valuable to both the instructor and your classmates. You are expected to attend class regularly and will need to do so to learn the material needed to meet expectations for assignments. You will be assigned relevant readings and practice activities and our class discussions will often rely on knowledge from these. We also will have in-class activities from time to time and you will receive credit for participating in these.

We will always be respectful of the opinions and voices of others in our classroom. As a class, we will develop a more specific set standards for what engagement means in our course, and you will have the opportunity to both evaluate and receive feedback on your engagement regularly.

Quizzes (240 points total/24%)

There will be 5 quizzes. Only 3 will count towards your final grade. The first quiz, Quiz 0, is a syllabus quiz that is untimed and online and counts for 80 points. All of the other quizzes will be closed book, written on paper, and you will be prohibited from having any accessible electronic devices. However, you may bring a 3x5 notecard for each quiz with any information you would like. You will turn in your notecard with the quiz. The instructor will have notecards available during the semester so you do not need to purchase them. The quizzes will be designed to take ~20-30 minutes but you will have the entire class period.

There are no make-ups for quizzes. The 2 drops cover any case in which you would request a make-up. Do not use your drops haphazardly.

Students with university-approved and documented accommodations can use the documented accommodations afforded to them. **Be sure to speak to me about your accommodations and how they apply to our course – they are ALWAYS honored.**

NOTE: Assignments will either be submitted on paper in class or on Blackboard. Emailed assignments will not be accepted.

Taskmasters (240 points total/24%):

Taskmasters will be problem-solving scenarios assigned in class on Mondays which you will turn in on Friday. Creativity and discussion is strongly encouraged! The goal will be to come up with processes and protocols to use AI for problem-solving. There will be 5 Taskmasters assigned but only 4 will count towards your final grade. This means you can drop 1 Taskmaster if you do not complete it. **The drop covers any case in which you would ask for an extension.**

Taskmasters will be turned in on paper – you will write/type them up and hand them in at the beginning of class - with a 10 min. grace period. This gives those of you coming from across campus a few extra minutes.

Homework Assignments (250 points total/25%):

There are four homework assignments including Homework 0 which is the introduction assignment (EASY FREE POINTS!). The number of points assigned to each homework is variable. Homeworks will have some components submitted on Blackboard and some submitted **handwritten at the beginning of class (with the 10 minute grace period).**

Can I work in a group?

Yes, you are always encouraged to work in a group with your classmates of up to 4 students. You are welcome to turn in a single assignment for the group (though I'd recommend having at least 2 people submit it in case someone forgets). If you choose to work in a group, read the late hours policy below carefully.

Can I turn my homework in late? Can I have an extension?

To cover cases of illness, technological issues, and any other circumstances under which you might request an extension, you are afforded late blocks and passes. These are set up in a way to model the sick and/or personal leave you might acquire in a job…but in some sense simplified to help us both keep track of them but also rather complex due to the various modalities for assignments. You have a total of **10 late blocks** and **3 late passes** for the semester. The policies governing the use of these are as follows: READ CAREFULLY

- **You do not need to ask or provide a reason to use late blocks/passes.** Your privacy is respected.
- Late blocks /passes cannot be used on quizzes and Taskmasters. Late blocks are for things turned in electronically, late passes are for things turned in in class. They are not interchangeable.
- **For components turned in on Blackboard (LATE BLOCKS): 1 late BLOCK = 6 hours.** They are not divisible in smaller increments. Increments less than 6 hours (e.g. 1 minute) will be rounded up.
- **For components turned in on paper in class (LATE PASS): 1 late PASS= 1 class period.** This means 1 late pass allows you to turn in the on-paper work at the beginning (THE BEGINNING) of the next class period. This is not divisible in smaller increments. I can guarantee I can receive your assignment in

class, do not try to turn it in by any other means other than personally handing it to me in class. If you turn in your work at the end of the following class period, that is **2 late passes used**.

- Do not request to turn in an assignment late and NOT use late blocks/passes. **If you have none remaining, you may turn the assignment component (Blackboard or written) in up to 2 days/next class period late for up to 60% credit.** After 2 days, you can no longer turn in the assignment.

- **If you work in a group, your late blocks/passes do not cover other individuals in the group.** This means if you are responsible for turning the assignment, and you forget and turn it in a day late and one of your groupmates has no late blocks, they will only be able to obtain 60% credit.

- Anyone with documented accommodations from the university may be subject to revised policies– **TALK TO ME and we will establish guidelines that respect your accommodations**.

Final Project (150 points total/15%):

The Final Project will be a chance for you to maximize your creativity and show off your problem-solving skills. We will have a two project milestones during the semester to help you avoid getting stuck and/or being overwhelmed. The milestones will be worth 50 points total and you will get credit just for doing them with some effort on time. The first milestone will be a topic submission (10 points) and the second will be to participate in an informal design session in class in which you present your ideas(40 points).

The Final Project due date and time assigned for our course is Tuesday December 8 1700 (this is the later date/time for the 2 sections).

Important information on Final Exam Scheduling (link to forms/info: https://www.wm.edu/offices/studentsuccess/academic-wellbeing/forms/final-exam-schedule-adjustment-request.php))

- If you have multiple exams consecutively or an exam that overlaps, provide documentation to the Office of Academic Advising by the LAST DAY OF CLASSES
- If you are unable to submit your project on the assigned date due to extenuating circumstances, you will need to contact Academic Wellbeing

Final Grade Determination

Final grades will be assigned as follows:

92%-100%: A	90%-<92%: A-	
87% -<90%: B+	82%-<87%: B	80%-<82%: B-
77%-<80%:C+	72%-<77%:C	70%-<72%: C-
67%-<70%:D+	60-<67%:D	<60%: F

Grades within 0.5 will be rounded. Grades more than 0.5 away will not be rounded. If a grade cutoff is adjusted (e.g. a B+ becomes 86) that will be done for all students. Awarding grade bumps upon request is unethical, which in turn makes requesting a grade bump from the instructor unethical. Emails requesting grade bumps will not receive an answer.

What is the AI policy? What are the other collaboration policies?

You are encouraged to collaborate with your classmates on assignments. You should not be collaborating with people not in the class/friends/family members etc.

Our class is about AI, so you will need to use AI in some capacity for the out of class assignments. However, the goal is for you to be the manager and architect of AI solutions....not for AI to do the solving for you. You will find that you achieve better solutions when you work collaboratively with AI rather than just submitting an assignment wholesale as a prompt and copying what it produces without even thinking about it.

BE AWARE: You may be asked to explain your submitted work in detail at some point. You should be prepared to do so.

What about accommodations? And military service? How do those fit in with course policies?

It is the policy of William & Mary to accommodate students with disabilities and qualifying diagnosed conditions in accordance with federal and state laws. Any student who feels they may need an accommodation based on the impact of a learning, psychiatric, physical or chronic health diagnosis should be referred to Student Accessibility Services staff at 757-221-2509 or at sas@wm.edu. SAS staff will work with you to determine if accommodations are warranted, and if so, to help you obtain an official letter of accommodation. Accommodations afforded to you by SAS will always be honored. **Be sure to speak with the instructor if you have any questions about how your accommodations will apply in the course.**

Military service is an excused absence from course attendance and necessitates creation of an alternative completion schedule for course work. If the dates of your service will overlap with the dates of our course, please speak to the instructor in advance so we can organize a suitable course plan.

Additional Information

William & Mary recognizes that students juggle different responsibilities and can face challenges that make learning difficult. There are many resources available at W&M to help students navigate emotional/psychological, physical/medical, material/accessibility concerns, including: The W&M Counseling Center at (757) 221-3620. Services are free and confidential; The W&M Health Center at (757) 221-4386. For additional support or resources & questions, contact the Dean of Students at 757-221-2510.
Click the link for a list of other resources available to students